

# A Cheatsheet for Lie Algebra

These notes follow Humphreys, *Introduction to Lie Algebras and Representation Theory* [1].

## 1 Basic Concepts

### 1.1 Preliminaries

**Definition 1.1.** A vector space  $L$  over a field  $F$  with an operation  $L \times L \rightarrow L$   
 $(x, y) \mapsto [xy]$   
called the bracket or commutator of  $x, y$  is called a Lie algebra over  $F$  if the following axioms are satisfied  $\left\{ \begin{array}{l} \text{the bracket operator is bilinear} \\ \forall x \in L, [xx] = 0 \text{ where the last} \\ \forall x, y, z \in L, [x[yz]] + [y[zx]] + [z[xy]] = 0 \end{array} \right.$   
condition is called the Jacobi identity.

$$\begin{aligned} [x+y, x+y] &= [x, x] + [x, y] + [y, x] + [y, y] && \text{by the linearity} \\ \rightarrow &= [x, y] + [y, x] && [x, x] = 0, \forall x \in L \Leftrightarrow [x, y] = -[y, x] \\ &= 0 && [x, x] = 0, \forall x \in L \end{aligned}$$

and if  $\text{char } F \neq 2$ , the converse also holds.

**Definition 1.2.** Two Lie algebras  $L, L'$  over  $F$  are isomorphic if there exists a vector space isomorphism  $\phi : L \rightarrow L'$  satisfying  $\phi([xy]) = [\phi(x)\phi(y)]$ ,  $\forall x, y \in L$ .

**Definition 1.3.** A subalgebra of a Lie algebra  $L$  is a subspace  $K$  of  $L$  s.t.  $[xy] \in K$ ,  $\forall x, y \in K$ .

$\rightarrow K$  is a Lie algebra with the inherited operations.

e.g. **The General Linear Algebra**  $\mathfrak{gl}(V)$

If  $V$  is a finite dimensional vector space over  $F$ , the set of linear transformation  $V \rightarrow V$  is denoted by  $\text{End } V$ , which is of dimension  $n^2$  where  $\dim V = n$  and is a ring relative to the usual product operation. Define a new operation  $[x, y] = xy - yx$ , called the bracket of  $x, y$ . With this operation,  $\text{End } V$  becomes a Lie algebra over  $F$ .

$\rightarrow \begin{cases} \forall x \in L, [x, x] = xx - xx = 0 \\ \forall x, y, z \in L, [x[yz]] + [y[zx]] + [z[xy]] = x(yz - zy) + y(zx - xz) + z(xy - yx) = 0 \end{cases}$   
 This Lie algebra is called the general linear algebra  $\mathfrak{gl}(V)$ . It is identified with the set of all  $n \times n$  matrices over  $F$ , denoted  $\mathfrak{gl}(n, F)$ . Then  $[e_{ij}, e_{kl}] = \delta_{jk}e_{il} - \delta_{li}e_{kj}$  where  $e_{ij}$  is a standard basis for matrices.

### Subalgebras of the General Linear Algebra

- **the Special Linear Algebra** is the set of endomorphisms of  $V$  having trace zero and is denoted  $\mathfrak{sl}(V) = \mathfrak{sl}(l+1, F)$  where  $\dim V = l+1$ .

Throughout,  $T(\cdot)$  denotes the trace. Since  $\begin{cases} T(xy) = T(yx) \\ T(x+y) = T(x) + T(y) \end{cases}$ , it is a subalgebra of  $\mathfrak{gl}(V)$ .

Take all  $e_{ij}$ ,  $i \neq j$  along with  $h_i = e_{ii} - e_{i+1, i+1}$ . Then this is the basis for  $\mathfrak{sl}(V)$  and therefore it has dimension  $l + (l+1)^2 - (l+1)$ .

- **the symplectic algebra** is the set of endomorphisms  $x$  of  $V$  satisfying  $f(x(v), w) = -f(v, x(w))$ , denoted  $\mathfrak{sp}(V) = \mathfrak{sp}(2l, F)$ , where  $\dim V = 2l$  and  $f$  has a skew-symmetric form  $s = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$ . Then for  $x = \begin{bmatrix} m & n \\ p & q \end{bmatrix}$  where  $m, n, p, q \in$

$\mathfrak{gl}(l, F)$ ,  $sx = -x's$  given it being symplectic. In other words,  $\begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix} \begin{bmatrix} m & n \\ p & q \end{bmatrix} =$

$$-\begin{bmatrix} m' & p' \\ n' & q' \end{bmatrix} \Leftrightarrow \begin{cases} n' = n \\ p' = p \\ m' = -q \end{cases}. \text{ Then the basis consists of}$$

$$\begin{cases} m' = -q \\ n = n' \end{cases} \begin{cases} e_{ii} - e_{l+i, l+i}, 1 \leq i \leq l \\ e_{ij} - e_{l+j, l+i} \\ e_{i, l+i} \\ e_{i, l+j} + e_{j, l+i} \end{cases} \begin{matrix} l + l^2 - l \\ l + \frac{1}{2}(l(l-1)) \end{matrix}. \text{ Since } n \text{ and } q \text{ share}$$

the basis, the total dimension becomes  $2l^2 + l$ .

- **the Orthogonal Algebra** is the set of endomorphisms  $x$  of  $V$  satisfying  $f(x(v), w) = -f(v, x(w))$ , denoted  $\mathfrak{o}(V) = \mathfrak{o}(2l, F)$  where  $\dim V = 2l$  and  $f$  has a form  $s = \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix}$ . Then for  $x = \begin{bmatrix} m & n \\ p & q \end{bmatrix}$  where  $m, n, p, q \in \mathfrak{gl}(l, F)$ ,  $sx = -x's$

given it being orthogonal. In other words,  $\begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix} \begin{bmatrix} m & n \\ p & q \end{bmatrix} = -\begin{bmatrix} m' & p' \\ n' & q' \end{bmatrix} \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix}$ .

Then the basis consists of

$$\begin{cases} m' = -q \\ n = n' \end{cases} \begin{cases} \text{the diagonal contributes nothing, since } m' = -q \text{ forces } m_{ii} = 0 \\ e_{ij} - e_{l+j, l+i}, i \neq j \\ e_{i, l+i} \\ e_{i, l+j} + e_{j, l+i} \end{cases} \begin{matrix} l^2 - l \\ l + \frac{1}{2}(l(l-1)) \end{matrix}$$

Since  $n$  and  $q$  share the basis, the total dimension becomes  $2l^2 - l$ .

- $\left\{ \begin{array}{ll} \text{the set of upper triangular matrices } (a_{ij}), a_{ij} = 0, \forall i > j & \mathfrak{t}(n, F) \\ \text{the set of strictly upper triangular matrices } (a_{ij}), a_{ij} = 0, \forall i \geq j & \mathfrak{n}(n, F) \\ \text{the set of all diagonal matrices} & \mathfrak{d}(n, F) \end{array} \right. \rightarrow$
- $\mathfrak{t}(n, F) = \mathfrak{n}(n, F) \oplus \mathfrak{d}(n, F)$   
 pf. The basis for  $\mathfrak{t}(n, F)$  consists of all  $e_{ij}$  s.t.  $i \leq j$  which has dimension  $\frac{1}{2}n(n+1)$ . The basis for  $\mathfrak{d}(n, F)$  consists of all  $e_{ii}$  which has dimension  $n$ , while the basis for  $\mathfrak{n}(n, F)$  consists of all  $e_{ij}$  s.t.  $i < j$  which has dimension  $\frac{1}{2}n(n-1)$ . Then the sum of bases for  $\mathfrak{n}(n, F)$ ,  $\mathfrak{d}(n, F)$  gives the basis for  $\mathfrak{t}(n, F)$  and the stmt follows. Moreover the sum is direct:  $\mathfrak{n}(n, F)$  is nonabelian while  $\mathfrak{d}(n, F)$  is abelian, and their bases are disjoint ( $e_{ij}$  with  $i < j$  versus  $e_{ii}$ ), so  $\mathfrak{n}(n, F) \cap \mathfrak{d}(n, F) = 0$  and  $\mathfrak{t}(n, F) = \mathfrak{n}(n, F) \oplus \mathfrak{d}(n, F)$ .

**Definition 1.4.** For any field  $F$ , an  $F$ -algebra is a vector space  $\mathcal{U}$  over  $F$  endowed with a bilinear operation  $\mathcal{U} \times \mathcal{U} \rightarrow \mathcal{U}$ , usually denoted by juxtaposition. By a derivation of  $\mathcal{U}$ , we mean a linear map  $\delta : \mathcal{U} \rightarrow \mathcal{U}$  satisfying the product rule  $\delta(ab) = a\delta(b) + \delta(a)b$ .

Since a Lie algebra  $L$  is an  $F$ -algebra in the above sense,  $\text{Der}L$  is defined. If  $x \in L$ ,  $y \mapsto [xy]$  is an endomorphism of  $L$ , denoted  $\text{adx}$ . Then  $\text{adx} \in \text{Der}L$  by the Jacobi identity. Derivations of this form are called **inner**, all others **outer**.

**Definition 1.5.** The map  $\begin{array}{ccc} L & \rightarrow & \text{Der}L \\ x & \mapsto & \text{adx} \end{array}$  is called the adjoint representation of  $L$ .

**Definition 1.6.** A Lie algebra  $L$  over  $F$  is called abelian if  $\forall x, y \in L$ ,  $[xy] = 0$ .

**Definition 1.7.** A subspace  $I$  of a Lie algebra  $L$  is called an ideal of  $L$  if  $x \in L$ ,  $y \in I$  imply  $[xy] \in I$ .

$\rightarrow [yx] = -[xy] \in I$  by skew-symmetry.

e.g.  $\left\{ \begin{array}{l} Z(L) = \{z \in L \mid [xz] = 0, \forall x \in L\}, \text{ the centre of } L \\ [LL] \text{ the derived algebra of } L \end{array} \right.$

Then  $L$  is abelian iff  $[LL] = 0$ . At the other extreme, when  $L = \mathfrak{sl}(n, F)$ ,  $[LL] = L$ .

If  $I, J$  are two ideals of a Lie algebra  $L$ , then  $\left\{ \begin{array}{l} I + J = \{x + y \mid x \in I, y \in J\} \\ [IJ] = \{\sum [x_i y_i] \mid x_i \in I, y_i \in J\} \end{array} \right.$  are ideals.

**Definition 1.8.** A Lie algebra  $L$  is called simple if it has no ideals except itself and 0, and  $[LL] \neq 0$ .

e.g.  $L = \mathfrak{sl}(2, F)$ ,  $\text{char} F \neq 2$ . Then it has a standard basis  $x = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ ,  $y = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$ ,  $h = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ . Then the multiplication table is determined by the equations  $\begin{cases} [xy] = h \\ [hx] = 2x \\ [hy] = -2y \end{cases}$ . If  $I \neq 0$  is an ideal of  $L$ , let  $ax + by + ch$  be an arbitrary

nonzero element of  $I$ . Then applying  $\begin{cases} \text{adx twice, we get } -2bx \in I \\ \text{ady twice, we get } -2ay \in I \end{cases}$ . If  $a$  or  $b$  is nonzero, that means either  $x$  or  $y$  is in  $I$ , and therefore  $L = I$  by the above multiplication table. If  $a = b = 0$ ,  $ch \in I$  and this also implies  $L = I$  by the above multiplication table. This shows  $L$  is simple.

If  $L$  is not simple, it is possible to factor out a nonzero proper ideal  $I$  and obtain a **quotient algebra**  $L/I$  : as vector space  $L/I$  is just the quotient space, while its Lie multiplication is defined by  $[x + I, y + I] = [xy] + I$ .

**Definition 1.9.** For any Lie algebras  $L, L'$  over  $F$ , a linear transformation  $\phi : L \rightarrow L'$  is called a *homomorphism* if  $\phi([xy]) = [\phi(x)\phi(y)]$ ,  $\forall x, y \in L$ . In addition, if  $\ker \phi = 0$ ,  $\phi$  is called a *monomorphism*, or if  $\text{Im} \phi = L'$ ,  $\phi$  is called an *epimorphism*. It is an *isomorphism* if it is both mono- and epi-.

**$\ker \phi$  is an ideal of  $L$ .**

pf.  $\forall x \in \ker \phi$ ,  $\forall y \in L$ ,  $\phi([xy]) = [\phi(x)\phi(y)] = 0$  so  $[xy] \in \ker \phi$  and therefore the stmt holds.

**$\text{Im} \phi$  is a subalgebra of  $L'$ .**

**Proposition 1.1.** 1. If  $\phi : L \rightarrow L'$  is a homomorphism of Lie algebras, then  $L/\ker \phi \cong \text{Im} \phi$ . If  $I$  is any ideal of  $L$  included in  $\ker \phi$ , there exists a unique

homomorphism  $\psi : L/I \rightarrow L'$  making the following diagram commute:

$$\begin{array}{ccc} L & \xrightarrow{\phi} & L' \\ & \searrow \pi & \uparrow \psi \\ & & L/I \end{array}$$

2. If  $I, J$  are ideals of  $L$  s.t.  $I \subset J$ , then  $J/I$  is an ideal of  $L/I$  and  $(L/I)/(J/I)$  is naturally isomorphic to  $L/J$ .

3. If  $I, J$  are ideals of  $L$ , there is a natural isomorphism between  $(I + J)/J$  and  $I/(I \cap J)$ .

**Definition 1.10.** A representation of a Lie algebra  $L$  is a homomorphism  $\phi : L \rightarrow \mathfrak{gl}(V)$ .

For  $L$  finite-dimensional, we have an adjoint representation  $\text{ad} : L \rightarrow \mathfrak{gl}(L)$ . Then its kernel consists of all  $x \in L$  for which  $\text{adx} = 0$ , i.e.  $[xy] = 0$ . So  $\ker \text{ad} = Z(L)$ . Then, if  $L$  is simple,  $Z(L) = 0$  so that  $\text{ad}L : L \rightarrow \mathfrak{gl}(L)$  is a monomorphism. This means that any simple Lie algebra is isomorphic to a linear Lie algebra.

**Definition 1.11.** An automorphism of  $L$  is an isomorphism of  $L$  onto itself and it is denoted  $\text{Aut}L$ .

If  $g \in \text{GL}(V)$  is any invertible endomorphism of  $V$  and if moreover  $gLg^{-1} = L$ , then  $x \mapsto gxg^{-1}$  is an automorphism of  $L$ .

Assume  $\text{char}F = 0$ . Suppose  $x \in L$  is an element for which  $\text{adx}$  is nilpotent, i.e.  $(\text{adx})^k = 0$ ,  $\exists k \in \mathbb{Z}_{++}$ . Then  $\exp(\text{adx}) = 1 + \text{adx} + \frac{(\text{adx})^2}{2!} + \dots + \frac{(\text{adx})^{k-1}}{(k-1)!}$ .

**CLAIM :**  $\exp(\text{adx}) \in \text{Aut}L$ .

$$\begin{aligned} [\exp \delta(x), \exp \delta(y)] &= \left[ \sum_{i=0}^{k-1} \frac{\delta^i x}{i!}, \sum_{j=0}^{k-1} \frac{\delta^j y}{j!} \right] \\ &= \sum_{n=0}^{2k-2} \left( \sum_{i=0}^n \left[ \frac{\delta^i x}{i!}, \frac{\delta^{n-i} y}{(n-i)!} \right] \right) \\ \text{pf.} \quad &= \sum_{n=0}^{2k-2} \frac{\delta^n [xy]}{n!} \quad \frac{\delta^n}{n!} [xy] = \sum_{i=0}^n \left[ \frac{\delta^i x}{i!}, \frac{\delta^{n-i} y}{(n-i)!} \right] \quad \text{where } \text{adx} \\ &= \sum_{n=0}^{k-1} \frac{\delta^n [xy]}{n!} \quad \delta^k = 0 \\ &= \exp \delta([xy]) \end{aligned}$$

is replaced by an arbitrary nilpotent derivation  $\delta$  of  $L$ . Also given  $\exp \delta = 1 + \eta$ , its inverse is  $1 - \eta + \eta^2 - \dots \pm \eta^{k-1}$  exists and therefore  $\exp \delta$  is invertible. This gives the claim.

**Definition 1.12.** An automorphism of the form  $\exp(\text{adx})$ ,  $\text{adx}$  nilpotent is called inner and the subgroup of  $\text{Aut}L$  generated by these is denoted  $\text{Int}L$ , the set of inner automorphisms.

$\rightarrow$  it is a normal subgroup (  $\phi \in \text{Aut}L$ ,  $x \in L$ ,  $\phi(\text{adx})\phi^{-1} = \text{ad}\phi(x) \rightarrow \phi \exp(\text{adx})\phi^{-1} = \exp(\text{ad}\phi(x))$  ).

**CLAIM :** If  $L \subset \mathfrak{gl}(V)$  is an arbitrary linear Lie algebra, and  $x \in L$  is nilpotent, then  $(\exp x)y(\exp x)^{-1} = \exp \text{ad}(x)(y)$ ,  $\forall y \in L$ .

pf.  $\text{adx} = \lambda_x + \rho_{-x}$  where  $\begin{cases} \lambda_x \text{ left multiplication by } x \\ \rho_{-x} \text{ right multiplication by } -x \end{cases} \in \text{End}(\text{End}V)$ . Then  $\exp \text{adx} = \exp(\lambda_x + \rho_{-x}) = \exp \lambda_x \exp \rho_{-x} = \lambda_{\exp x} \cdot \rho_{\exp -x}$  and the stmt follows.

**Definition 1.13.** Define a sequence of ideals of  $L$  called the derived series by  $L^{(0)} = L$ ,  $L^{(1)} = [LL]$ ,  $L^{(2)} = [L^{(1)}L^{(1)}]$ ,  $\dots$ ,  $L^{(i)} = [L^{(i-1)}L^{(i-1)}]$ . Then  $L$  is said to be solvable if  $L^{(n)} = 0$ ,  $\exists n \in \mathbb{Z}_{++}$ .

e.g.  $\mathfrak{t}(n, F)$  is solvable.

pf. The basis for it consists of the matrix units  $e_{ij}$ ,  $i \leq j$  and the dimension is

$\frac{n(n+1)}{2}$ . Since  $[e_{ii}e_{il}] = e_{il}$ ,  $\forall i < l$ ,  $\mathfrak{n}(n, F) \subset [LL]$  where it is the subalgebra of upper triangular nilpotent matrices. Since  $\mathfrak{t}(n, F) = \mathfrak{d}(n, F) + \mathfrak{n}(n, F)$  and  $\mathfrak{d}(n, F)$  is abelian,  $\mathfrak{n}(n, F)$  is equal to the derived algebra of  $L$ . Define the level of  $e_{ij}$  by  $j - i$ . In the formula of commutators, assume  $i < j, k < l$ . Wlog, assume  $i \neq l$ . Then  $[e_{ij}e_{kl}] = \begin{cases} e_{il} & j = k \\ 0 & \text{otherwise} \end{cases}$ . Each  $e_{il}$  is a commutator of two matrices whose levels add up to that of  $e_{il}$ . Thus  $L^{(2)}$  is spanned by  $e_{ij}$  of level  $\geq 2$ ,  $L^{(i)}$  by those of level  $\geq 2^{i-1}$ . Since  $L^{(i)} = 0$  whenever  $2^{i-1} > n - 1$  and therefore the stmt follows.

**Proposition 1.2.** *Let  $L$  be a Lie algebra.*

1. *If  $L$  is solvable, then so are all subalgebras and homomorphic images of  $L$ .*
2. *If  $I$  is a solvable ideal of  $L$  s.t.  $L/I$  is solvable, then  $L$  itself is solvable.*
3. *If  $I, J$  are solvable ideals of  $L$ , then so is  $I + J$ .*

pf.

- If  $K$  is a subalgebra of  $L$ ,  $K^{(i)} \subset L^{(i)}$  and therefore the subalgebras are also solvable. If  $\phi : L \rightarrow M$  is an epimorphism, an easy induction on  $i$  shows  $\phi(L^{(i)}) = M^{(i)}$  and therefore its homomorphic images are also solvable.
- Say  $(L/I)^{(n)} = 0$ . For the canonical homomorphism  $\pi : L \rightarrow L/I$ , 1 implies  $\pi(L^{(n)}) = 0$ , i.e.  $L^{(n)} \subset I = \ker \pi$ . By hypothesis,  $\exists m \in \mathbb{Z}_{++}$  s.t.  $I^{(m)} = 0 \rightarrow (L^{(n)})^{(m)} = L^{(n+m)} = 0$  so 2 follows.
- There exists an isomorphism between  $(I + J)/J$  and  $I/(I \cap J)$ . As a homomorphic image of  $I$ , the RHS is solvable so  $(I + J)/J$  is solvable. Then so is  $I + J$  by 2.

→let  $L$  be an arbitrary Lie algebra and let  $S$  be a maximal solvable ideal. If  $I$  is any other solvable ideal of  $L$ , by the previous proposition 3,  $S + I = S$  by maximality, hence  $I \subset S$ . Thus there exists a unique maximal solvable ideal of  $L$ , called the **radical** of  $L$ , denoted  $\text{Rad}L$ . If  $\text{Rad}L = 0$ ,  $L$  is called **semisimple**.

## 1.2 Engel's Theorem

**Definition 1.14.** *Define a sequence of ideals of  $L$  called the descending central series by  $L^0 = L, L^1 = [LL], \dots, L^i = [LL^{i-1}]$ .  $L$  is called **nilpotent** if  $\exists n \in \mathbb{Z}_{++}$  s.t.  $L^n = 0$ .*

→  $L^{(i)} \subset L^i$ ,  $\forall i$  so nilpotent algebras are solvable but not vice versa.

pf.  $L = \mathfrak{t}(n, F)$  is solvable. But  $L^{(1)} = L^1 = \mathfrak{n}(n, F) = [LL^1] = L^2 = \dots \forall i \geq 1$ . Thus it is not nilpotent.

**Proposition 1.3.** *Let  $L$  be a Lie algebra.*

1. *If  $L$  is nilpotent, then so are all subalgebras and homomorphic images of  $L$ .*
2. *If  $L/Z(L)$  is nilpotent, then so is  $L$ .*
3. *If  $L$  is nilpotent, and nonzero, then  $Z(L) \neq 0$ .*

pf.

- by the previous proposition, part 1
- $L^n \subset Z(L)$  implies  $L^{n+1} = [LL^n] \subset [LZ(L)] = 0$
- for the descending central series,  $\exists n \in \mathbb{Z}_{++}$  s.t.  $L^{n-1} = [LL^{n-2}] \neq 0, L^n = [LL^{n-1}] = 0$ . This means  $0 \neq L^{n-1} \subset Z(L)$  and therefore  $Z(L) \neq 0$ .

$\rightarrow \exists n \in \mathbb{Z}_{++}$  s.t.  $\text{adx}_1 \dots \text{adx}_n(y) = 0, \forall x_i, y \in L$  implies  $L$  to be nilpotent.

**Definition 1.15.** *If  $L$  is any Lie algebra and  $x \in L$ , we call  $x$  ad-nilpotent if  $\text{adx}$  is a nilpotent endomorphism.*

**Lemma 1.1.** *Let  $x \in \mathfrak{gl}(V)$  be a nilpotent endomorphism. Then  $\text{adx}$  is also nilpotent.*

pf. Associate two endomorphisms  $\begin{cases} \lambda_x & \text{left translation} \\ \rho_x & \text{right translation} \end{cases}$  to  $x$ . Then they are nilpotent since  $x$  is. Moreover  $\lambda_x \rho_x(y) = xyx = \rho_x \lambda_x(y)$ , i.e. they commute. In any ring the sum or difference of two commuting nilpotents is again nilpotent since  $x, y$  nilpotent with  $m, n$  s.t.  $x^m = 0 = y^n, (x \pm y)^{m+n} = 0$ . So  $\text{adx} = \lambda_x - \rho_x$  is nilpotent.

**Theorem 1.1.** *Let  $L$  be a subalgebra of  $\mathfrak{gl}(V)$ ,  $V$  finite dimensional. If  $L$  consists of nilpotent endomorphisms and  $V \neq 0$ , then there exists nonzero  $v \in V$  for which  $Lv = 0$ .*

pf.

- If  $\dim L = 0$ , the theorem is trivial.
- If  $\dim L = 1$ , given  $L$  is nilpotent  $Lv = 0$ .
- If  $\dim L > 1$ ,  $\exists K \subsetneq L$  proper subalgebra of  $L$ . By the previous lemma,  $K$  acts via  $\text{ad}$  as a Lie algebra of nilpotent linear transformation on the vector space  $L$ , hence also on the vector space  $L/K$ . Because  $\dim K < \dim L$ , by the inductive hypothesis,  $\exists x \in L$  with  $x + K \neq K$  such that  $x + K$  is killed by the image of  $K$  in  $\mathfrak{gl}(L/K)$ . Then  $\forall y \in K, [yx] \in K$  while  $x \notin K$ , i.e.  $K$  is properly

included in  $N_L(K)$ . Take  $K$  to be a maximal proper subalgebra of  $L$ . By the previous argument,  $N_L(K) = L$ , i.e.  $K$  is an ideal of  $L$ . If  $\dim L/K$  were greater than 1, then the inverse image in  $L$  of one dimensional subalgebra of  $L/K$  would be a proper subalgebra properly containing  $K$ , a contradiction. Thus  $K$  has codimension 1. Thus  $L = K + Fz$ ,  $z \in L \setminus K$ . By induction,  $W = \{v \in V \mid Kv = 0\}$  is nonzero. Since  $K$  is an ideal,  $W$  is stable under  $L$ , i.e.  $\forall x \in L, \forall y \in K, \forall w \in W, yxw = xyw - [xy]w = 0$ . Choose  $z \in L \setminus K$  as above, so the nilpotent endomorphism  $z$  has an eigenvector, i.e.  $\exists v \in W$  for which  $zv = 0$ . Then  $Lv = 0$  as desired.

**Theorem 1.2** (Engel). *If all elements of  $L$  are ad-nilpotent, then  $L$  is nilpotent.*

pf. For any Lie algebra  $L$  all of whose elements are ad-nilpotent, the algebra  $\text{ad}L \subset \mathfrak{gl}(L)$  satisfies the hypothesis of the previous theorem. Thus  $\exists 0 \neq x \in L$  s.t.  $[Lx] = 0$ , i.e.  $Z(L) \neq 0$ . Now  $L/Z(L)$  consists of ad-nilpotent elements and has smaller dimension than  $L$ . By inductive hypothesis,  $L/Z(L)$  is nilpotent. The previous proposition, part 2, then implies that  $L$  itself is nilpotent.

**Definition 1.16.** *If  $V$  is a finite dimensional vector space, a flag in  $V$  is a chain of subspaces  $0 = V_0 \subsetneq V_1 \subsetneq \cdots \subsetneq V_n = V$ ,  $\dim V_i = i$ . If  $x \in \text{End}V$  we say  $x$  stabilises this flag provided  $xV_i \subset V_i$ ,  $\forall i$ .*

**Corollary 1.1.** *Under the hypotheses of the theorem, there exists a flag  $(V_i)$  in  $V$  stable under  $L$  with  $xV_i \subset V_{i-1}$ ,  $\forall i$ .*

pf. By Theorem 1.0.1, there exists  $0 \neq v \in V$  s.t.  $Lv = 0$ . Set  $V_1 = Fv$  and let  $W = V/V_1$ . Then the induced action of  $L$  on  $W$  is also by nilpotent endomorphisms. By induction on  $\dim V$ ,  $W$  has a flag stabilised by  $L$ , whose inverse image in  $V$  does the trick.

**Lemma 1.2.** *Let  $L$  be nilpotent,  $K$  an ideal of  $L$ . Then if  $K \neq 0$ ,  $K \cap Z(L) \neq 0$ .*

pf.  $L$  acts on  $K$  via the adjoint representation, so by Theorem 1.0.1,  $\exists x \in K$  s.t.  $[Lx] = 0$  so  $x \in K \cap Z(L)$ .

**CLAIM : Let  $L$  be nilpotent,  $K$  a proper subalgebra of  $L$ . Then  $N_L(K)$  includes  $K$  properly.**

pf. By Lemma 1.0.1,  $K$  acts via  $\text{ad}$  as a Lie algebra of nilpotent linear transformations on the vector space  $L$ , hence also on the vector space  $L/K$ . By Theorem 1.0.1,  $\exists x' \in L/K$  s.t.  $Kx' = 0$ . This implies  $\forall y \in K, [yx] \in K, x \in L \setminus K$ . Thus  $x \in N_L(K) \setminus K$  and therefore it is properly contained.

**CLAIM : Let  $L$  be nilpotent. Then  $L$  has an ideal of codimension 1.**

pf. Let  $K$  be a maximal proper subalgebra of  $L$ . By the previous claim,  $N_L(K) = L$ . Then  $[Ly] \subset K$ ,  $\forall y \in K$  and therefore  $K$  is an ideal of  $L$ . If  $\dim L/K$  were greater than 1, the inverse image in  $L$  of a one dimensional subalgebra of  $L/K$  would be a proper subalgebra properly containing  $K$ , a contradiction.

### 1.3 Theorems of Lie and Cartan

Assume throughout  $\text{char} F = 0$ .

**Theorem 1.3.** *Let  $L$  be a solvable subalgebra of  $\mathfrak{gl}(V)$ ,  $V$  finite dimensional. If  $V \neq 0$ , then  $V$  contains a common eigenvector for all the endomorphisms in  $L$ .*

pf.

- $\dim L = 0$ , it holds trivially
- $\dim L = 1$ , its proper subalgebra  $[LL] = 0$  and  $L$  is abelian of dimension 1 and an eigenvector for a basis vector of  $L$  gives the theorem.
- $\dim L > 1$ , since  $L$  is solvable,  $L$  properly includes  $[LL]$ .  $L/[LL]$  being abelian, any subspace is automatically an ideal. Take a subspace of codimension 1, then its inverse image  $K$  is an ideal of codimension 1 in  $L$ . By the inductive hypothesis,  $\exists v \in V$ ,  $\exists \lambda : K \rightarrow F$  s.t.  $xv = \lambda(x)v$ ,  $\forall x \in K$ . Fix this  $\lambda$  and set  $W = \{w \in V | xw = \lambda(x)w, \forall x \in K\} \neq 0$ .

– **CLAIM :  $L$  leaves  $W$  invariant.**

pf.  $\forall w \in W$ ,  $x \in L$ ,  $\forall y \in K$ ,  $yxw = xyw - [xy]w = \lambda(y)xw - \lambda([xy])w = \lambda(y)(xw)$  iff  $\lambda([xy]) = 0$ . So we only need to prove  $\lambda([xy]) = 0$ . Fix  $w \in W$ ,  $x \in L$ . Let  $n \in \mathbb{Z}_{++}$  be the smallest integer for which  $w, xw, \dots, x^n w$  are linearly dependent. Let  $W_i$  be the subspace of  $V$  spanned by  $w, xw, \dots, x^{i-1}w$ , so  $\dim W_n = n$ ,  $W_n = W_{n+i}$ ,  $\forall i \geq 0$  and  $x$  maps  $W_n$  into  $W_n$ . Relative to the basis  $w, xw, \dots, x^{n-1}w$  of  $W_n$ ,

$$\begin{aligned} yx^i w &= yxx^{i-1}w = xyx^{i-1}w - [xy]x^{i-1}w \\ &= x\lambda(y)x^{i-1}w + xw' - [xy]x^{i-1}w \quad \text{by the inductive hypothesis,} \\ &\equiv \lambda(y)x^i w \pmod{W_i} \quad x \text{ maps } W_{i-1} \text{ into } W_i \end{aligned}$$

where  $w' \in W_{i-1}$ . This means  $y \in K$  is represented by an upper triangular matrix whose diagonal entries are equal to  $\lambda(y)$ . Then  $T|_{W_n}(y) = n\lambda(y)$ . In particular, this is true for elements of  $K$  of the special form  $[xy]$ . But  $x, y$  both stabilise  $W_n$ , so  $[xy]$  acts on  $W_n$  as the commutator of two endomorphisms of  $W_n$  and therefore its trace is 0. So  $n\lambda([xy]) = 0$ . Given  $\text{char} F = 0$ ,  $\lambda([xy]) = 0$ .

Then write  $L = K + Fz$ . Since  $F$  is algebraically closed,  $\exists v_0 \in W$  an eigenvector of  $z$ . Then  $v_0$  is obviously a common eigenvector for  $L$ .

**Corollary 1.2 (Lie).** *Let  $L$  be a solvable subalgebra of  $\mathfrak{gl}(V)$ ,  $\dim V = n < \infty$ . Then  $L$  stabilises some flag in  $V$ .*

pf. By the previous theorem with induction on  $\dim V$ .

**Corollary 1.3.** *Let  $L$  be solvable. Then there exists a chain of ideals of  $L$ ,  $0 = L_0 \subsetneq \dots \subsetneq L_n = L$  s.t.  $\dim L_i = i$ .*

**Corollary 1.4.** *Let  $L$  be solvable. Then  $x \in [LL]$  implies  $\text{ad}_L x$  is nilpotent. In particular,  $[LL]$  is nilpotent.*

pf. Find a flag of ideals in the previous corollary. Relative to a basis  $(x_1, \dots, x_n)$  of  $L$  for which  $(x_1, \dots, x_i)$  spans  $L_i$ , the matrices of  $\text{ad}_L$  lie in  $\mathfrak{t}(n, F)$ . Thus the matrices of  $[\text{ad}_L, \text{ad}_L] = \text{ad}_L[LL] \subset \mathfrak{n}(n, F)$  the derived algebra of  $\mathfrak{t}(n, F)$ . Thus  $\text{ad}_L x$  is nilpotent for  $x \in [LL]$ .  $\text{ad}_{[LL]} x$  is nilpotent, so  $[LL]$  is nilpotent by Engel's theorem.

**Definition 1.17.**  $x \in \text{End} V$  is semisimple iff the roots of its minimal polynomial over  $F$  are all distinct.

$\Leftrightarrow$  given  $F$  being algebraically closed,  $x$  is semisimple iff  $x$  is diagonalisable.

**Proposition 1.4.** *Let  $V$  be a finite dimensional vector space over  $F$ ,  $x \in \text{End} V$ .*

1. *There exists unique  $x_s, x_n \in \text{End} V$  satisfying the conditions;  $x = x_s + x_n$ ,  $\begin{cases} x_s & \text{semisimple} \\ x_n & \text{nilpotent} \end{cases}$  which commute.*
2. *There exist polynomials  $p(T), q(T)$  in one indeterminate, without constant term, s.t.  $x_s = p(x)$ ,  $x_n = q(x)$ . In particular,  $x_s, x_n$  commute with any endomorphism commuting with  $x$ .*
3. *If  $A \subset B \subset V$  are subspaces, and  $x$  maps  $B$  into  $A$ , then  $x_s, x_n$  also map  $B$  into  $A$ .*

pf. Let  $a_1, \dots, a_l$  be the distinct eigenvalues of  $x$ , so the characteristic polynomial is  $\prod (T - a_i)^{m_i}$  where  $m_i$  denote the multiplicity. If  $V_i = \ker(x - a_i \cdot 1)^{m_i}$ ,  $V$  is the direct sum of the subspaces  $V_1, \dots, V_l$  each stable under  $x$ . On  $V_i$ ,  $x$  clearly has characteristic polynomial  $(T - a_i)^{m_i}$ . By CRT, we get  $p(T) \equiv \begin{cases} a_i & \text{mod } (T - a_i)^{m_i} \\ 0 & \text{mod } T \end{cases}$ . Set  $q(T) = T - p(T)$ . Then  $p(T), q(T)$  has zero constant term since  $p(T) \equiv 0 \pmod T$ .

- 2,3

Set  $\begin{cases} x_s = p(x) \\ x_n = q(x) \end{cases}$ . Since they are polynomials in  $x$ , they commute with each other as well as all endomorphisms which commute with  $x$ .

They stabilise all subspaces of  $V$  stabilised by  $x$ , in particular the spaces  $V_i$ . The congruence  $p(T) \equiv a_i \pmod{(T-a_i)^{m_i}}$  shows the restriction of  $x_s - a_i \cdot 1$  to  $V_i$  is zero for all  $i$ , hence  $x_s$  acts diagonally on  $V_i$  with single eigenvalue  $a_i$ . By definition,  $x_n = x - x_s$  which makes it clear that  $x_n$  is nilpotent. Because  $p(T), q(T)$  have no constant term, 3 follows.

- 1

Existence follows by the previous argument. If there exists  $x = s + n$  another decomposition,  $x_s - s = n - x_n$ . Then by 2, all endomorphisms in the equation commute. Sums of commuting semisimple/nilpotent endomorphisms are semisimple/nilpotent, whereas only 0 can be both semisimple and nilpotent. Thus  $x_s = s$ ,  $x_n = n$ .

→ the decomposition  $x = x_s + x_n$  is called the **Jordan-Chevalley decomposition** of  $x$ .

Consider the adjoint representation of the Lie algebra  $\mathfrak{gl}(V)$ ,  $V$  finite dimensional. If  $x \in \mathfrak{gl}(V)$  is nilpotent, then so is  $\text{adx}$  by the lemma 1.2.1. Similarly, **if  $x$  is semisimple, then so is  $\text{adx}$** .

pf. Choose a basis  $(v_1, \dots, v_n)$  of  $V$  relative to which  $x$  has matrix  $\text{diag}(a_1, \dots, a_n)$ . Let  $\{e_{ij}\}$  be the standard basis of  $\mathfrak{gl}(V)$  relative to  $(v_1, \dots, v_n)$ . Then  $e_{ij}(v_k) = \delta_{jk}v_i$ . Then  $\text{adx}(e_{ij}) = (a_i - a_j)e_{ij}$  so  $\text{adx}$  has diagonal matrix, relative to the chosen basis of  $\mathfrak{gl}(V)$ .

**Lemma 1.3.** *Let  $x \in \text{End}V$ ,  $x = x_s + x_n$  its Jordan decomposition. Then  $\text{adx} = \text{adx}_s + \text{adx}_n$  is the Jordan decomposition of  $\text{adx}$ .*

pf. Since  $\text{adx}_s, \text{adx}_n$  are semisimple and nilpotent respectively, they commute, since  $[\text{adx}_s, \text{adx}_n] = \text{ad}[x_s, x_n] = 0$ . Thus by the previous proposition 1, the lemma follows.

**Lemma 1.4.** *Let  $\mathcal{U}$  be a finite dimensional  $F$ -algebra. Then  $\text{Der}\mathcal{U}$  contains the semisimple and nilpotent parts in  $\text{End}\mathcal{U}$  of all its elements.*

pf. If  $\delta \in \text{Der}\mathcal{U}$ , let  $\sigma, \nu \in \text{End}\mathcal{U}$  be its semisimple and nilpotent parts respectively. If  $a \in F$ , set  $\mathcal{U}_a = \{x \in \mathcal{U} \mid (\delta - a1)^k x = 0, \exists k\}$ . Then  $\mathcal{U}$  is the direct sum of those  $\mathcal{U}_a$  for which  $a$  is an eigenvalue of  $\delta$  and  $\sigma$  acts on  $\mathcal{U}_a$  as scalar multiplication by  $a$ . For arbitrary  $a, b \in F$ ,  $\mathcal{U}_a \mathcal{U}_b \subset \mathcal{U}_{a+b}$  since  $(\delta - (a+b)1)^n(xy) =$

$\sum_{i=0}^n \binom{n}{i} ((\delta - a1)^{n-i}x) \cdot ((\delta - b1)^i y)$ ,  $\forall x, y \in \mathcal{U}$ . If  $x \in \mathcal{U}_a$ ,  $y \in \mathcal{U}_b$ ,  $\sigma(xy) = (a+b)xy$  because  $xy \in \mathcal{U}_{a+b}$ . On the other hand,  $\sigma xy + x\sigma y = (a+b)xy$ . By directness of the sum  $\mathcal{U} = \bigoplus \mathcal{U}_a$ ,  $\sigma \in \text{Der } \mathcal{U}$ . Then  $\nu = \delta - \sigma \in \text{Der } \mathcal{U}$ .

**Lemma 1.5.** *Let  $A \subset B$  be two subspaces of  $\mathfrak{gl}(V)$ ,  $\dim V < \infty$ . Set  $M = \{x \in \mathfrak{gl}(V) \mid [xB] \subset A\}$ . Suppose  $x \in M$  satisfies  $T(xy) = 0$ ,  $\forall y \in M$ . Then  $x$  is nilpotent.*

pf. Let  $x = s + n$  be the Jordan decomposition of  $x$ . Fix a basis  $v_1, \dots, v_m$  of  $V$  relative to which  $s$  has matrix  $\text{diag}(a_1, \dots, a_m)$ . Let  $E$  be the vector subspace of  $F$  over the prime field  $\mathbb{Q}$  spanned by the eigenvalues  $a_1, \dots, a_m$ . Consider the dual space  $E^*$ , whose elements are the linear functions  $f : E \rightarrow \mathbb{Q}$ . Given such  $f \in E^*$ , let  $y$  be any element of  $\mathfrak{gl}(V)$  whose matrix relative to the given basis is  $\text{diag}(f(a_1), \dots, f(a_m))$ . If  $\{e_{ij}\}$  is the corresponding basis of  $\mathfrak{gl}(V)$ ,  $\begin{cases} \text{ads}(e_{ij}) = (a_i - a_j)e_{ij} \\ \text{ady}(e_{ij}) = (f(a_i) - f(a_j))e_{ij} \end{cases}$ . Let  $r(T) \in F[T]$  be a polynomial without constant term satisfying  $r(a_i - a_j) = f(a_i) - f(a_j)$  for all pairs  $i, j$ . The existence of such  $r(T)$  follows from Lagrange interpolation. Then  $\text{ady} = r(\text{ads})$ . Since  $\text{ads}$  is the semisimple part of  $\text{adx}$  by the lemma 1.3.1, it can be written as a polynomial in  $\text{adx}$  without constant term by the proposition 1.3.1. Therefore  $\text{ady} = r(\text{ads})$  is also a polynomial in  $\text{adx}$  without constant term. By the hypothesis of the lemma,  $\text{adx}$  maps  $B$  into  $A$ , so  $\text{ady}(B) \subset A$  i.e.  $y \in M$ . By the hypothesis of the lemma,  $T(xy) = 0 \rightarrow \sum a_i f(a_i) = 0$ . The LHS is a  $\mathbb{Q}$ -linear combination of elements of  $E$ . Applying  $f$  to the above equation gives  $\sum (f(a_i))^2 = 0$ . But the numbers  $f(a_i)$  are rational, so this forces all of them to be 0, and therefore  $f$  must be 0 because the  $a_i$  span  $E$ . Then  $E^*$  is zero and its dual  $E$  is also 0 and the lemma follows.

**Lemma 1.6.**  *$T([xy]z) = T(x[yz])$  for all endomorphisms  $x, y, z$  of a finite-dimensional vector space.*

pf.  $\begin{cases} [xy]z = xyz - yxz \rightarrow T([xy]z) = T(xyz) - T(yxz) \\ x[yz] = xyz - xzy \rightarrow T(x[yz]) = T(xyz) - T(xzy) \end{cases}$  are equivalent given  $T(ab) = T(ba)$ .

**Theorem 1.4** (Cartan's Criterion). *Let  $L$  be a subalgebra of  $\mathfrak{gl}(V)$ ,  $V$  finite dimensional. Suppose that  $T(xy) = 0$ ,  $\forall x \in [LL]$ ,  $y \in L$ . Then  $L$  is solvable.*

pf. Let  $\begin{cases} A = [LL] \\ B = L \end{cases}$ , so  $M = \{x \in \mathfrak{gl}(V) \mid [xL] \subset [LL]\}$ . Then  $L \subset M$  trivially. By the hypothesis  $T(xy) = 0$ ,  $\forall x \in [LL]$ ,  $\forall y \in L$ . If  $[xy]$  is a typical generator of  $[LL]$ , and if  $z \in M$ , the equation above shows  $T([xy]z) = T(x[yz]) = T([yz]x)$ . By the definition of  $M$ ,  $[yz] \in [LL]$  so  $T(xy) = 0$ ,  $\forall x \in [LL]$ ,  $\forall y \in M$ . Then by the previous lemma, every  $x \in [LL]$  is nilpotent. Then by Engel's theorem,  $[LL]$  is nilpotent and therefore  $L$  is solvable.

## 2 Semisimple Algebras

### 2.1 Killing Form

**Definition 2.1.** Let  $L$  be any Lie algebra. If  $x, y \in L$ , define the killing form  $\kappa(x, y) = T(\text{adxady})$ .

→ it is symmetric bilinear and associative in the sense that  $\kappa([xy], z) = \kappa(x, [yz])$ .

**Lemma 2.1.** Let  $I$  be an ideal of  $L$ . If  $\kappa$  is the killing form of  $L$  and  $\kappa_I$  be the killing form of  $I$ , then  $\kappa_I = \kappa|_{I \times I}$ .

pf. If  $W$  is a subspace of a finite dimensional vector space  $V$ , and  $\phi$  an endomorphism of  $V$  mapping  $V$  into  $W$ , then  $T\phi = T\phi|_W$ , since we can extend a basis of  $W$  to a basis of  $V$  and calculate  $\phi$ . Then if  $x, y \in I$ ,  $\text{adxady}$  is an endomorphism of  $L$ , mapping  $L$  into  $I$ , so its trace  $\kappa(x, y)$  coincides with the trace  $\kappa_I(x, y)$  of  $\text{adxady}|_I = \text{ad}_I x \text{ad}_I y$ .

**Definition 2.2.** A symmetric bilinear form  $\beta(x, y)$  is called nondegenerate if its radical  $S$  is 0, where  $S = \{x \in L | \beta(x, y) = 0, \forall y \in L\}$ .

→  $S$  is an ideal of  $L$ , given the associativity of the killing form.

**Theorem 2.1.** Let  $L$  be a Lie algebra. Then  $L$  is semisimple iff its killing form is nondegenerate.

pf.

- Assume  $L$  is semisimple. Then  $\text{Rad}L = 0$ . Let  $S$  be the radical of  $\kappa$ . By definition,  $T(\text{adxady}) = 0, \forall x \in S, \forall y \in L$ . By the Cartan criterion,  $\text{ad}_L S$  is solvable, hence  $S$  is solvable. Since  $S$  is an ideal of  $L$ ,  $S \subset \text{Rad}L = 0$  and  $\kappa$  is nondegenerate.
- Let  $S = 0$  and  $I$  be an abelian ideal of  $L$ . Suppose  $x \in I, y \in L$ . Then  $\text{adxady}$  maps  $L \rightarrow L \rightarrow I$  and  $(\text{adxady})^2$  maps  $L$  into  $[II] = 0$ . This means  $\text{adxady}$  is nilpotent, hence  $0 = T(\text{adxady}) = \kappa(x, y)$  so  $I \subset S = 0$ .

→  $S \subset \text{Rad}L$  but not vice versa.

pf. Let  $L$  be a Lie algebra with basis  $x, y, z$  s.t. 
$$\begin{cases} [xy] = z \\ [xz] = -y \\ [yz] = 0 \end{cases} \text{ . Then } \text{adx} =$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}, \text{ady} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}, \text{ad}(z) = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ and therefore its } \kappa = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

So  $\text{Rad}L = L \not\subset \text{span}(y, z) = S \equiv \text{radical of } \kappa$ .

**Definition 2.3.** A Lie algebra  $L$  is said to be the direct sum of ideals  $I_1, \dots, I_t$  provided  $L = I_1 + \dots + I_t$ , denoted  $L = I_1 \oplus \dots \oplus I_t$ .

$$\rightarrow [I_i I_j] \subset I_i \cap I_j = 0, \forall i \neq j.$$

**Theorem 2.2.** Let  $L$  be semisimple. Then there exist ideals  $L_1, \dots, L_t$  of  $L$  which are simple s.t.  $L = L_1 \oplus \dots \oplus L_t$ . Every simple ideal of  $L$  coincides with one of the  $L_i$ . Moreover, the killing form of  $L_i$  is the restriction of  $\kappa$  to  $L_i \times L_i$ .

pf.

- **CLAIM : Let  $I$  be an arbitrary ideal of  $L$ . Then  $L = I \oplus I^\perp$ .**

pf.  $I^\perp = \{x \in L : \kappa(x, y) = 0, \forall y \in I\}$  is also an ideal by the associativity of  $\kappa$ . By Cartan's criterion applied to  $I \cap I^\perp$ , it is solvable, hence 0 as an ideal of the semisimple  $L$ . Thus since  $\dim I + \dim I^\perp = \dim L$ ,  $L = I \oplus I^\perp$ .

- Existence

If  $L$  has no proper nonzero ideal, then  $L$  itself is simple and there is nothing to prove. Otherwise, let  $L_1$  be a minimal nonzero ideal. Then by the first claim,  $L = L_1 \oplus L_1^\perp$ . In particular, any ideal of  $L_1$  is also an ideal of  $L$  and therefore  $L_1$  is semisimple, and by minimality it is simple. For the same reason  $L_1^\perp$  is semisimple. By inductive hypothesis, it splits into a direct sum of simple ideals, which are also ideals of  $L$ .

- Uniqueness

If  $I$  is any simple ideal of  $L$ , then  $[IL]$  is also an ideal of  $L$ , nonzero because  $Z(L) = 0$ . Then  $[IL] = I$  by simplicity. On the other hand,  $[IL] = [IL_1] + \dots + [IL_t]$  so all but one summand must be 0. Say  $[IL_i] = I$ . Then  $I \subset L_i$ , which forces  $I = L_i$  by the simplicity of  $L_i$ .

**Corollary 2.1.** If  $L$  is semisimple,  $L = [LL]$  and all ideals and homomorphic images of  $L$  are semisimple. Moreover, each ideal of  $L$  is a sum of certain simple ideals of  $L$ .

**For any Lie algebra  $L$ ,  $\text{ad}L$  is an ideal in  $\text{Der}L$ .**

pf.  $\forall y, [\delta, \text{adx}](y) = \delta \text{adx}(y) - \text{adx} \delta(y) = \delta([xy]) - [x, \delta y] = [\delta x, y] + [x, \delta y] - [x, \delta y] = [\delta x, y] = \text{ad}(\delta x)(y) \Leftrightarrow [\delta, \text{adx}] = \text{ad}(\delta x), \forall x \in L, \forall \delta \in \text{Der}L$ . So the stmt follows.

**Theorem 2.3.** If  $L$  is semisimple, then  $\text{ad}L = \text{Der}L$ .

pf. Since  $L$  is semisimple,  $Z(L) = 0$ . Thus  $L \rightarrow \text{ad}L$  is an isomorphism of Lie algebras. In particular,  $M = \text{ad}L$  has nondegenerate killing form by the theorem 1.4.1. If  $D = \text{Der}L$ ,  $[DM] \subset M$ . Then by the lemma 1.4.1,  $\kappa_M$  is the restriction to  $M \times M$  of the killing form  $\kappa_D$  of  $D$ . In particular, if  $I = M^\perp$  is the subspace of  $D$  orthogonal to  $M$  under  $\kappa_D$ , then the nondegeneracy of  $\kappa_M$  forces  $I \cap M = 0$ . Both  $I, M$  are ideals of  $D$  so  $[IM] = 0$ . If  $\delta \in I$ ,  $\text{ad}\delta x = 0$ ,  $\forall x \in L$  by the previous stmt, so  $\delta x = 0$ ,  $\forall x \in L$  because  $\text{ad}$  is one to one and  $\delta = 0$ . So  $I = 0$ ,  $\text{Der}L = M = \text{ad}L$ .

→ For any semisimple Lie algebra  $L$ , if  $\mathcal{U}$  is any  $F$ -algebra of finite dimension,  $\text{Der}\mathcal{U}$  contains the semisimple and nilpotent parts in  $\text{End}\mathcal{U}$  of all its elements. In particular, since  $\text{Der}L$  coincides with  $\text{ad}L$  while  $L \rightarrow \text{ad}L$  is one to one, each  $x \in L$  determines unique elements  $s, n \in L$  s.t.  $\text{ad}x = \text{ad}s + \text{ad}n$  is the Jordan decomposition of  $\text{ad}x$ . This means  $x = s + n$ ,  $[sn] = 0$ ,  $\begin{cases} s & \text{ad-semisimple} \\ n & \text{ad-nilpotent} \end{cases}$  and  $\begin{cases} s = x_s & \text{the semisimple part} \\ n = x_n & \text{the nilpotent part} \end{cases}$  of  $x$ .

## 2.2 Complete Reducibility of Representations

**Definition 2.4.** Let  $L$  be a Lie algebra. A vector space  $V$ , endowed with an operation  $\begin{matrix} L \times V & \rightarrow & V \\ (x, v) & \mapsto & xv \end{matrix}$  is called an  $L$ -module if the following conditions are satisfied:  $\begin{cases} (ax + by)v = a(xv) + b(yv), x(av + bw) = a(xv) + b(xw) \\ [xy]v = xyv - yxv \end{cases}, \forall x, y \in L. \forall v, w \in V, a, b \in F.$

e.g. If  $\phi : L \rightarrow \text{gl}(V)$  is a representation of  $L$ , then  $V$  may be viewed as an  $L$ -module via the action  $xv = \phi(x)(v)$ . Conversely, given an  $L$ -module  $V$ , this equation defines a representation  $\phi : L \rightarrow \text{gl}(V)$ .

**Definition 2.5.** A homomorphism of  $L$ -modules is a linear map  $\phi : V \rightarrow W$  s.t.  $\phi(xv) = x\phi(v)$ . When  $\phi$  is an isomorphism of vector spaces, we call it an isomorphism of  $L$ -modules.

**Definition 2.6.** An  $L$ -module  $V$  is called irreducible if it has precisely two  $L$ -submodules- itself and  $0$ .

**Definition 2.7.**  $V$  is called completely reducible if  $V$  is a direct sum of irreducible  $L$ -submodules or equivalently if each  $L$ -submodule  $W$  of  $V$  has a complement  $W'$ .

→ if it is a direct sum of irreducible  $L$ -submodules, its any submodule  $W$  is also a direct sum of irreducible  $L$ -submodules. Then the summand that does not belong to its submodule also makes a  $L$  submodule  $W^\perp$  s.t.  $W \cap W^\perp = 0$ ,  $W + W^\perp = L$

so  $L = W \oplus W^\perp$ . Conversely, if any submodule of  $L$  has its complement, pick a minimal submodule of  $L$ . If there is no proper submodule,  $L$  is irreducible and there is nothing to prove. Otherwise, pick a minimal proper submodule  $L_1$  of  $L$ . Then by minimality, it is irreducible. Then it has complement  $L_1^\perp$  with  $\dim L_1^\perp < \dim L$ . Then by the inductive hypothesis it is a direct sum of irreducible  $L$ -submodules and therefore  $L = L_1 \oplus L_1^\perp$  is.

Given a representation  $\phi : L \rightarrow \mathfrak{gl}(V)$ , the associative algebra generated by  $\phi(L)$  in  $\text{End}V$  leaves invariant precisely the same subspaces as  $L$ . Therefore, all the usual results for modules over associative rings hold for  $L$  as well.

**SCHUR'S LEMMA : Let  $\phi : L \rightarrow \mathfrak{gl}(V)$  be irreducible. Then the only endomorphisms of  $V$  commuting with all  $\phi(x)$  are the scalars.**

$L$  itself is an  $L$ -module, and then an  $L$ -submodule is just an ideal, so a simple algebra  $L$  is irreducible as  $L$ -module, while a semisimple algebra is completely reducible.

## MAKING NEW MODULES out of OLD ONES

- dual module

Let  $V$  be a  $L$ -module. Then the dual vector space  $V^*$  becomes an  $L$ -module with  $(xf)(v) = -f(xv)$ ,  $\forall f \in V^*$ . Then  $([xy]f)(v) = -f([xy]v) = -f(xyv - yxv) = -f(xyv) + f(yxv) = (xf)(yv) - (yf)(xv) = -(yx f)(v) + (xy f)(v) = ((xy - yx)f)(v)$ .

- tensor-product

If  $V, W$  are  $L$ -modules, let  $V \otimes W$  be the tensor product over  $F$  of the underlying vector spaces. For Lie algebras, define  $x(v \otimes w) = xv \otimes w + v \otimes xw$ . Then  $[xy](v \otimes w) = [xy]v \otimes w + v \otimes [xy]w = (xyv - yxv) \otimes w + v \otimes (xyw - yxw) = (xyv \otimes w + v \otimes xyw) - (yxv \otimes w + v \otimes yxw) = (xy - yx)(v \otimes w)$ .

→ given a vector space  $V$  over  $F$ ,  $\exists$  an isomorphism 
$$\begin{array}{ccc} V^* \otimes V & \rightarrow & \text{End}V \\ f \otimes v & \mapsto & (w \mapsto f(w)v) \end{array}$$
 enables us to consider  $\text{End}V$  as an  $L$ -module via the isomorphism. The action of  $L$  on  $\text{End}V$  is  $(x \cdot \psi)(v) = x\psi(v) - \psi(xv)$ ,  $\forall \psi \in \text{End}V$ ,  $\forall v \in V$ .

Let  $L$  be semisimple and  $\phi : L \rightarrow \mathfrak{gl}(V)$  be a faithful representation of  $L$ . Define a symmetric bilinear form  $\beta(x, y) = T(\phi(x)\phi(y))$  on  $L$ . The form  $\beta$  is associative, so its radical  $S$  is an ideal of  $L$ . Moreover  $\phi(S) \cong S$  is solvable, so  $S = 0$  and  $\beta$  is non-degenerate. Let  $L$  be semisimple,  $\beta$  any nondegenerate symmetric associative bilinear form on  $L$ . If  $(x_1, \dots, x_n)$  is a basis of  $L$ , there is a uniquely determined dual basis  $(y_1, \dots, y_n)$  relative to  $\beta$  satisfying  $\beta(x_i, y_j) = \delta_{ij}$ . If  $x \in L$ , 
$$\begin{cases} [xx_i] = \sum_j a_{ij} x_j \\ [xy_i] = \sum_j b_{ij} y_j \end{cases}.$$

$$\begin{aligned}
a_{ik} &= \sum_j a_{ij} \beta(x_j, y_k) = \beta([xx_i], y_k) && \text{by the definition of } \beta \\
&= \beta(-[x_i x], y_k) && \text{by the skew-symmetry of bracket} \\
\text{Then } &= \beta(x_i, -[xy_k]) && \text{by the associativity of } \beta \\
&= -\sum_k b_{kj} \beta(x_i, y_j) && = -b_{ki}
\end{aligned}$$

$L \rightarrow \mathfrak{gl}(V)$  is any representation of  $L$ , write  $c_\phi(\beta) = \sum_i \phi(x_i) \phi(y_i) \in \text{End} V$  where

$$\begin{aligned}
[\phi(x) c_\phi(\beta)] &= \sum_i [\phi(x) \phi(x_i)] \phi(y_i) + \sum_i \phi(x_i) [\phi(x) \phi(y_i)] \\
x_i, y_i \text{ run over dual bases relative to } \beta. \text{ Then } &= \sum_{ij} a_{ij} \phi(x_j) \phi(y_i) + \sum_{ij} b_{ij} \phi(x_i) \phi(y_j) \\
&= 0
\end{aligned}$$

Thus  $c_\phi(\beta)$  is an endomorphism of  $V$  commuting with  $\phi(L)$ .

$\rightarrow c_\phi(\beta)$  is called the Casimir element of  $\phi$  and its trace is  $\sum_i T(\phi(x_i) \phi(y_i)) = \sum_i \beta(x_i, y_i) = \dim L$ . If  $\phi$  is also irreducible, by Schur's lemma, it should be scalar.

e.g.  $L = \mathfrak{sl}(2, F)$ ,  $V = F^2$ ,  $\phi$  is the identity map  $L \rightarrow \mathfrak{gl}(V)$ . Let  $(x, h, y)$  be the standard basis of  $L$ . Then its dual basis relative to the trace form is  $(y, h/2, x)$  so

$$c_\phi = xy + \frac{h^2}{2} + yx = \begin{bmatrix} 3/2 & 0 \\ 0 & 3/2 \end{bmatrix}.$$

When  $\phi$  is not faithful, we can factor out  $\ker \phi$  and get  $L'$  the sum of the remaining simple ideals after factoring out  $\ker \phi$ . Then  $\phi|_{L'}$  is faithful and we can get the Casimir element for  $L'$ . Then  $\phi(L) = \phi(L')$  commute with the Casimir element.

**Lemma 2.2.** *Let  $\phi : L \rightarrow \mathfrak{gl}(V)$  be a representation of a semisimple Lie algebra  $L$ . Then  $\phi(L) \subset \mathfrak{sl}(V)$ . In particular,  $L$  acts trivially on any one dimensional  $L$ -module.*

pf.  $L = [LL]$  and  $\mathfrak{sl}(V)$  is the derived algebra of  $\mathfrak{gl}(V)$ , the lemma follows.

**Theorem 2.4 (Weyl).** *Let  $\phi : L \rightarrow \mathfrak{gl}(V)$  be a finite dimensional representation of a semisimple Lie algebra. Then  $\phi$  is completely reducible.*

pf.

- $V$  has an  $L$ -submodule  $W$  of codimension 1

Since by the previous lemma,  $L$  acts trivially on  $V/W$ , wlog we may denote this module  $F$  and get a s.e.s.  $0 \rightarrow W \rightarrow V \rightarrow F \rightarrow 0$ .

– **CLAIM : wlog we can assume  $W$  is an irreducible  $L$ -module.**

Let  $W'$  be a proper nonzero submodule of  $W$ . Then we get another s.e.s.  $0 \rightarrow W/W' \rightarrow V/W' \rightarrow F \rightarrow 0$ . By the inductive hypothesis, it splits, so there is a one dimensional  $L$ -submodule of  $V/W'$  complementary to  $W/W'$ . Let it be  $\tilde{W}/W'$ . Then there exists another s.e.s  $0 \rightarrow$

$W' \rightarrow \tilde{W} \rightarrow F \rightarrow 0$  with  $\dim W' < \dim W$ , so by the inductive hypothesis, there exists a submodule  $X$  complementary to  $W'$ , i.e.  $\tilde{W} = W' \oplus X$ . But  $V/W' = W/W' \oplus \tilde{W}/W'$  so  $V = W \oplus X$ , given  $W \cap X = 0$  and dimension matching.

Let  $c = c_\phi$  be the Casimir element of  $\phi$ . Since  $c$  commutes with  $\phi(L)$ ,  $c$  is actually an  $L$ -module endomorphism of  $V$ . In particular,  $c(W) \subset W$  and  $\ker c$  is an  $L$ -submodule of  $V$ . Because  $L$  acts trivially on  $V/W$ ,  $c$  must do likewise as a linear combination of products of elements  $\phi(x)$ . So  $c$  has trace 0 on  $V/W$ . On the other hand,  $c$  acts as a scalar on the irreducible  $L$ -submodule  $W$  by the Schur's lemma. This cannot be 0, because that would force  $T|_V(c) = 0$ , a contradiction. Thus  $\ker c$  is a one-dimensional  $L$ -submodule of  $V$  which intersects  $W$  trivially, which is the desired complement of  $W$ .

- General case

Let  $W$  be a nonzero submodule of  $V$ . Then there exists a s.e.s.  $0 \rightarrow W \rightarrow V \rightarrow V/W \rightarrow 0$ . Let  $\text{hom}(V, W)$  be the space of linear maps  $V \rightarrow W$ , viewed as  $L$ -module. Let  $\mathcal{V}$  be the subspace of  $\text{hom}(V, W)$  consisting of those maps whose restriction to  $W$  is a scalar multiplication. Then  $f|_W = a1_W$ ,  $\forall x \in L$ ,  $\forall w \in W$ ,  $(xf)(w) = xf(w) - f(xw) = a(xw) - a(xw) = 0 \rightarrow (xf)|_W = 0$  so  $\mathcal{V}$  is an  $L$ -submodule. Let  $\mathcal{W}$  be the subspace of  $\mathcal{V}$  consisting of those  $f$  whose restriction to  $W$  is zero. Then  $\mathcal{W}$  is also an  $L$ -submodule and  $L$  maps  $\mathcal{V}$  to  $\mathcal{W}$ . Moreover  $\mathcal{V}/\mathcal{W}$  has dimension one because  $f \in \mathcal{V}$  is determined by the scalar  $f|_W \bmod \mathcal{W}$ . This gives the first situation. Then  $\mathcal{V}$  has a one-dimensional submodule complementary to  $\mathcal{W}$ . Let  $f : V \rightarrow W$  span it, so after multiplying by a nonzero scalar, we may assume  $f|_W = 1_W$ . Since  $f$  maps  $V$  into  $W$  and acts as  $1_W$  on  $W$ , we can say  $V = W \oplus \ker f$ .

**Theorem 2.5.** *Let  $L \subset \mathfrak{gl}(V)$  be a semisimple linear Lie algebra with  $V$  finite dimensional. Then  $L$  contains the semisimple and nilpotent parts in  $\mathfrak{gl}(V)$  of all its elements. In particular, the abstract and usual Jordan decompositions in  $L$  coincide.*

pf. Let  $x \in L$  arbitrary, with Jordan decomposition  $x = x_s + x_n$  in  $\mathfrak{gl}(V)$ . Since  $\text{adx}(L) \subset L$ ,  $\begin{cases} \text{adx}_s(L) \subset L \\ \text{adx}_n(L) \subset L \end{cases}$  where  $\text{ad} = \text{ad}_{\mathfrak{gl}(V)}$ , i.e.  $x_s, x_n \in N_{\mathfrak{gl}(V)}(L) = N$ , which is a Lie subalgebra of  $\mathfrak{gl}(V)$  including  $L$  as an ideal. If  $W$  is any  $L$ -submodule of  $V$ , define  $L_W = \{y \in \mathfrak{gl}(V) | y(W) \subset W, T(y|_W) = 0\}$ . For example,  $L_V = \mathfrak{sl}(V)$ . Since  $L = [LL]$ ,  $L$  lies in all such  $L_W$ . Set  $L' =$  intersection of  $N$  with all spaces  $L_W$ . Then  $L'$  is a subalgebra of  $N$  including  $L$  as an ideal. Then if  $x \in L$ ,  $x_s, x_n$  also lie in  $L_W$  and therefore in  $L'$ .

- **CLAIM :**  $L = L'$ .

pf.  $L'$  being finite dimensional  $L$ -module, by Weyl's theorem  $L' = L + M$  for some  $L$ -submodule  $M$ , where the sum is direct. But  $[LL'] \subset L$  given  $L' \subset N$  so the action of  $L$  on  $M$  is trivial. Let  $W$  be any irreducible  $L$ -submodule of  $V$ . If  $y \in M$ ,  $[Ly] = 0$  so by Schur's lemma,  $y$  acts on  $W$  as a scalar. On the other hand,  $T(y|_W) = 0$  because  $y \in L_W$ . Therefore  $y$  acts on  $W$  as zero.  $V$  can be written as a direct sum of irreducible  $L$ -submodules by Weyl's theorem so  $y = 0$ . Thus  $M = 0$ ,  $L = L'$ .

**Corollary 2.2.** *Let  $L$  be a semisimple Lie algebra,  $\phi : L \rightarrow \mathfrak{gl}(V)$  a finite dimensional representation of  $L$ . If  $x = s + n$  is the abstract Jordan decomposition of  $x \in L$ ,  $\phi(x) = \phi(s) + \phi(n)$  is the usual Jordan decomposition of  $\phi(x)$ .*

pf. The algebra  $\phi(L)$  is spanned by the eigenvectors of  $\text{ad}_{\phi(L)}\phi(s)$  since  $L$  has this property relative to  $\text{ads}$ , and therefore  $\text{ad}_{\phi(L)}\phi(s)$  is semisimple. Similarly,  $\text{ad}_{\phi(L)}\phi(n)$  is nilpotent and it commutes with  $\text{ad}_{\phi(L)}\phi(s)$ . Thus  $\phi(x) = \phi(s) + \phi(n)$  is the abstract Jordan decomposition of  $\phi(x)$  in the semisimple Lie algebra  $\phi(L)$ . Then by the previous theorem, the corollary follows.

**Definition 2.8.** *A Lie algebra  $L$  for which  $\text{Rad}L = Z(L)$  is called reductive.*

e.g.  $\left\{ \begin{array}{l} L \text{ abelian} \\ L = \mathfrak{gl}(n, F) \end{array} \right\}$ , then it is reductive.

- **If  $L$  is reductive,  $L$  is completely reducible  $\text{ad}L$ -module.**

pf. By the hypothesis  $L/Z(L)$  is semisimple. Since  $\text{ad}L$  acts trivially on  $Z(L)$ , the adjoint representation of  $L$  on  $L$  factors through  $L/Z(L)$ . Thus  $L$  is an  $L/Z(L)$ -module.  $\phi : L/Z(L) \rightarrow \mathfrak{gl}(V)$  is a representation of a semisimple Lie algebra. Then  $\phi$  is completely reducible by the Weyl's theorem. Thus  $L$  is completely reducible.

- **If  $L$  is a completely reducible  $\text{ad}L$ -module, then  $L$  is reductive.**

pf. There is an adjoint representation  $\phi : L \rightarrow \mathfrak{gl}(V)$ , let  $R = \text{Rad}L$  be its radical. Then since it is an ideal of  $L$ , it is an  $\text{ad}L$ -submodule of  $L$ . Then,  $L = R \oplus S$  by  $L$  being completely reducible. Because  $\text{ad}L$ -submodules of the adjoint module are ideals,  $S$  is also an ideal. Thus  $\left\{ \begin{array}{l} [RS] \subset R \\ [RS] \subseteq S \end{array} \right\}$ . Thus  $[RS] \subseteq R \cap S = 0 \rightarrow [RS] = 0$ . Since  $R$  is solvable,  $[RR] \subsetneq R$  unless  $R = 0$  and  $[RR]$  is also an  $\text{ad}L$ -submodule. By complete reducibility, choose an  $\text{ad}L$ -submodule  $T \subseteq R$  s.t.  $R = [RR] \oplus T$ . Given  $T$  an ideal,  $\left\{ \begin{array}{l} [RT] \subseteq T \\ [RT] \subseteq [RR] \end{array} \right\} \rightarrow$

$[RT] = 0$ ,  $R = [RR] \oplus T$ . Thus  $[RR] = [R[RR]]$ . If  $[RR] \neq 0$ ,  $R^{(1)} = R^{(2)} = \dots$ , a contradiction. Thus  $[RR] = 0$  and  $R \subseteq Z(L)$ . Since  $Z(L) \subseteq R$  the stmt follows.

- **If  $L$  is reductive, then all finite dimensional representations of  $L$  in which  $Z(L)$  is represented by semisimple endomorphisms are completely reducible.**

pf. By the first stmt, it is a completely reducible  $\text{ad}L$ -module and  $L = Z(L) \oplus [LL]$  where  $[LL]$  is semisimple. Let  $\phi : L \rightarrow \mathfrak{gl}(V)$  be finite dimensional with every element of  $Z(L)$  represented semisimply. By Weyl's theorem, it is completely reducible as  $[LL]$ -module.  $[Z(L), L] = 0$  implies  $[\phi(z)\phi(x)] = 0$ ,  $\forall z \in Z(L)$ ,  $\forall x \in [LL]$ . Since  $\phi(z)$  are commuting semisimple endomorphisms over an algebraically closed field they are simultaneously diagonalisable. Thus  $V = \bigoplus_{\lambda} V_{\lambda}$  where  $\forall V_{\lambda}$  is a common eigenspace for  $Z(L)$ . Since  $\phi(z)$  commute with  $\phi([LL])$ ,  $xV_{\lambda} \subseteq V_{\lambda}$ ,  $\forall x \in [LL]$ . So  $V_{\lambda}$  is an  $[LL]$ -module and by Weyl's theorem  $V_{\lambda} = \bigoplus_j V_{\lambda,j}$  where  $V_{\lambda,j}$  are irreducible  $[LL]$ -module. But  $Z(L)$  acts on all  $V_{\lambda,j}$  by the scalar  $\lambda(z)$ ,  $V_{\lambda,j}$  is actually an  $L$ -submodule. If it is a proper  $L$ -submodule, then it is an  $[LL]$ -submodule, contradicting irreducibility. Hence  $V = \bigoplus_{\lambda,j} V_{\lambda,j}$  is a decomposition into irreducible  $L$ -modules.

### 2.3 Representations of $\mathfrak{sl}(2, F)$

Let  $L$  denote  $\mathfrak{sl}(2, F)$  whose standard basis consists of  $x = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ ,  $y = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$ ,  $h = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ . Then  $[hx] = 2x$ ,  $[hy] = -2y$ ,  $[xy] = h$ .

Let  $V$  be an arbitrary  $L$ -module. Since  $h$  is semisimple,  $h$  acts diagonally on  $V$  by the previous corollary. Then  $V$  is decomposed as direct sum of eigenspaces  $V_{\lambda} = \{v \in V | hv = \lambda v\}$ ,  $\lambda \in F$ . Whenever  $V_{\lambda} \neq 0$ ,  $\lambda$  is called a weight of  $h$  in  $V$  and we call  $V_{\lambda}$  a weight space.

**Lemma 2.3.** If  $v \in V_{\lambda}$ ,  $\begin{cases} xv \in V_{\lambda+2} \\ yv \in V_{\lambda-2} \end{cases}$ .

pf.  $\begin{cases} h(xv) = [hx]v + xhv = 2xv + \lambda xv = (\lambda + 2)xv \\ h(yv) = [hy]v + yhv = -2yv + \lambda yv = (\lambda - 2)yv \end{cases}$   
 $\rightarrow x, y$  are represented by nilpotent endomorphisms of  $V$ .

Since  $\dim V < \infty$  and  $V = \bigoplus_{\lambda \in F} V_{\lambda}$  is direct, there must exist  $V_{\lambda} \neq 0$  s.t.  $V_{\lambda+2} = 0$ . For such  $\lambda$ , any nonzero vector in  $V_{\lambda}$  will be called a maximal vector of weight

$\lambda$ .

Assume  $V$  is an irreducible  $L$ -module. Choose a maximal vector, say  $v_0 \in V$  and set  $v_{-1} = 0$ ,  $v_i = \frac{1}{i!}y^i v_0$ ,  $\forall i \geq 0$ .

**Lemma 2.4.**  $\begin{cases} (i) & hv_i = (\lambda - 2i)v_i \\ (ii) & yv_i = (i+1)v_{i+1} \\ (iii) & xv_i = (\lambda - i + 1)v_{i-1} \end{cases}$

pf.

- for  $i = 0$ ,  $hv_0 = \lambda v_0$
- $i > 0$ ,  $hv_i = \frac{1}{i}hyv_{i-1} = \frac{1}{i}([hy]v_{i-1} + yhv_{i-1}) = \frac{1}{i}(-2yv_{i-1} + (\lambda - 2(i-1))yv_{i-1}) = \frac{1}{i}((\lambda - 2i)yv_{i-1}) = (\lambda - 2i)v_i$
- $(i+1)v_{i+1} = (i+1)\frac{1}{(i+1)!}y^{i+1}v_0 = yv_i$
- $i = 0 \rightarrow xv_0 = (\lambda + 1)v_{-1} = (\lambda + 1)0 = 0$   
 $ixv_i = xyv_{i-1} = [xy]v_{i-1} + yxv_{i-1}$  definition of bracket  
 $= hv_{i-1} + yxv_{i-1}$   
 $i > 0 \rightarrow = (\lambda - 2(i-1))v_{i-1} + (\lambda - i + 2)yv_{i-2}$  i and inductive hypothesis  
 $= (\lambda - 2i + 2)v_{i-1} + (i-1)(\lambda - i + 2)v_{i-1}$  by (ii)  
 $= i(\lambda - i + 1)v_{i-1}$

$\rightarrow$  (i) implies the nonzero  $v_i$  are all linearly independent. But  $\dim V < \infty$ . Let  $m$  be the smallest integer for which  $\begin{cases} v_m \neq 0 \\ v_{m+1} = 0 \end{cases}$ . Then the subspace of  $V$  with basis  $(v_0, \dots, v_m)$  is an  $L$ -submodule different from 0. Because  $V$  is irreducible, this subspace must be all of  $V$  and moreover relative to the ordered basis  $(v_0, \dots, v_m)$ , the matrices of the endomorphisms representing  $x, y, h$  can be written down explicitly. Also for  $i = m+1$ ,  $xv_i = 0$  in the LHS of (iii), whereas the RHS is  $(\lambda - m)v_m$ . Since  $v_m \neq 0$ ,  $\lambda = m$ , i.e. the weight of a maximal vector is a nonnegative integer. We call it the highest weight of  $V$ . Moreover each weight  $\mu$  occurs with multiplicity one by (i). In particular, since  $V$  determines  $\lambda$  uniquely, the maximal vector  $v_0$  is the only possible one in  $V$ .

**Theorem 2.6.** *Let  $V$  be an irreducible module for  $L = \mathfrak{sl}(2, F)$ .*

1. *Relative to  $h$ ,  $V$  is the direct sum of weight spaces  $V_\mu$ ,  $\mu = m, m-2, \dots, -(m-2), -m$  where  $m+1 = \dim V$  and  $\dim V_\mu = 1$  for each such  $\mu$ .*
2.  *$V$  has up to nonzero scalar multiples a unique maximal vector, whose weight is  $m$ .*
3. *The action of  $L$  on  $V$  is given explicitly by the above formulas, if the basis is chosen in the prescribed fashion. In particular, there exists at most one irreducible  $L$ -module up to isomorphism of each possible dimension  $m+1$ ,  $m \geq 0$ .*

pf. By the preceding paragraph.

**Corollary 2.3.** *Let  $V$  be any finite dimensional  $L$ -module,  $L = \mathfrak{sl}(2, F)$ . Then the eigenvalues of  $h$  on  $V$  are all integers and each occurs along with its negative. Moreover, in any decomposition of  $V$  into direct sum of irreducible submodules, the number of summands is precisely  $\dim V_0 + \dim V_1$ .*

pf. If  $V = 0$  then nothing to prove. If  $V \neq 0$ , by the Weyl's theorem, we can write  $V$  as direct sum of irreducible submodules. Then by the previous theorem, the first part of the corollary follows. Each irreducible  $L$ -module has a unique occurrence of either the weight 0 or else the weight 1, but not both so the last part of the corollary follows.

### Existence of the maximal vector

pf. Look at the subalgebra  $B$  spanned by  $h, x$ . It is a solvable subalgebra of  $\mathfrak{gl}(V)$  with  $\dim V < \infty$ . Since  $V \neq 0$ ,  $V$  contains a common eigenvector  $v$  for all the endomorphisms in  $B$  by the Lie's theorem ( $[BB] = Fx$ ,  $[FxFx] = 0$ ).

• **CLAIM :  $v$  is a maximal vector.**

pf. Since  $v$  is a common eigenvector for  $B$ ,  $\exists \lambda, \mu \in F$  s.t.  $\begin{cases} hv = \lambda v \\ xv = \mu v \end{cases}$ . Then  $2xv = [hx]v = hxv - xhv = h(\mu v) - x(\lambda v) = \mu\lambda v - \lambda\mu v = 0$ . Since  $\text{char} F = 0$ ,  $xv = 0$  so  $v$  is a maximal vector.

## References

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